

(Nonlinear) Time series Project

(June 2026)

Below you will find the instructions for your final project for the part of the nonlinear time series forecasting.

The report should be named “*nonlinear_project_<surname>*” and uploaded to the e-learn platform.

The deadline is Saturday, June 30th, 2026.

Project topic:

For the time series that corresponds to your name (see the appendix at the end of this paper) download from the *e-learning environment* the .txt file containing the time series you are asked to study, according to the following:

- i. Calculate the time delay for properly embedding the corresponding system attractor and apparently present the time series and the system attractor.
- ii. Calculate the correlation integrals and estimate the minimum embedding dimension and the fractal dimension for this system.
- iii. Calculate the maximal Lyapunov exponent.
- iv. Exclude the last value of your timeseries and predict it by utilizing nonlinear timeseries forecasting methods. Calculate the relative prediction error (%) and make a brief comment on this.

For all the necessary calculations and analyses you may use one of the following (also combined):

- the R environment,
- Matlab,
- Python programming language,

according to your soft skills.

In presenting the requested study in the form of a report, you are asked to include next to the description of the analysis you perform (graphs, tables etc.) and the emerging results, the relevant explanations and comments. Finally, it is desirable to include the code/commands for the calculations performed.

Any use of AI is permitted if this aims in helping you to form the code or finding the necessary sources' documentation for creating your project. In this case, you need to report the way you used it in each case.

Commenting on the conclusions is of great importance.

APPENDIX

Table of Timeseries for each student

.txt	Name	.txt	Name	.txt	Name	.txt	Name
w1	Ali-Oglou	w11	Kaptanis	w21	Pertsinidis	w31	Voudouris
w2	Belitsa	w12	Ketsentsidis	w22	Sergoudis	w32	Zachariadou
w3	Chrysochoidis	w13	Kotsonis	w23	Skenderoglou	w33	Zavaropoulos
w4	Dimitros	w14	Koukouvini	w24	Skordas	w34	Zisis
w5	Diomis	w15	Mavridou	w25	Skriapas	w35	Zografos
w6	Foudi	w16	Miaris	w26	Statis		
w7	Fragkiadaki	w17	Mpalaskas	w27	Tangiris		
w8	Giacobbe	w18	Nikolakoudi	w28	Tourliaridis		
w9	Kakavou	w19	Nikou	w29	Tsatalas		
w10	Kanoutas	w20	Papadopoulou	w30	Tsivelikas		

COMPLEMENTARY MATERIAL

Some useful R scripts are accompanying this project assignment to help you perform the requested analysis.

You need to download the corresponding pdf files describing the packages you use to study them so that you can properly use their custom functions.